

SILVER OAK UNIVERSITY

Silver Oak College of Computer Application

INNOVATIVE ASSIGNMENT – 1

CyberSafe: I4C Cyber Threat Intelligence & Digital Safety Portal

Responsible Technology Innovation Project

Course	Ethical and Legal Frameworks for Digital Systems
Course Code	MDC5011C
Programme	Integrated Master of Science – Information Technology
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Prototype: CyberSafe educational cybercrime awareness and digital safety portal

Abstract

CyberSafe is a responsible-technology prototype designed to convert cybercrime awareness into an interactive learning experience. The project is aligned with the Innovative Assignment brief, which requires students to study recent cyber incidents, identify cybercrime types, examine ethical issues, map relevant legal provisions, recommend preventive measures, and produce a practical creative output.

The project uses official Indian Cyber Crime Coordination Centre (I4C) advisories as the principal case-study source. Five documented scam patterns are examined: WhatsApp Web account renting, matrimonial-platform investment/crypto fraud, fake CAPTCHA jobs, digital arrest scams, and fake stock-market investment websites/apps. The analysis focuses on attack flow, social-engineering mechanisms, ethical concerns, legal awareness, and responsible digital behaviour.

The prototype is implemented as a client-side educational website using HTML5, CSS3 and Vanilla JavaScript. It includes case studies, threat filtering, a Scam Detector, a Cyber Law Guide, digital-ethics guidance, a safety checklist, incident-response steps, and links to official reporting resources. The site is explicitly positioned as an academic awareness tool rather than a government service, legal service, or professional fraud-detection system.

Problem Statement

Cybercrime increasingly combines technology with human manipulation. Victims may encounter fake identities, urgent financial demands, fraudulent employment offers, impersonation of authorities, malicious links or apps, and misleading investment opportunities. A static list of security tips may not sufficiently explain how these attacks work or how ethical and legal considerations connect to everyday digital decisions.

The problem addressed by CyberSafe is therefore: how can a student-oriented digital solution transform real cybercrime patterns into understandable, interactive, and ethically responsible learning material while avoiding collection of unnecessary personal information?

1. Objectives and Scope

Objectives

- Study recent I4C cybercrime advisories and extract practical attack patterns.
- Classify the selected incidents by major cybercrime/fraud mechanism.
- Explain ethical concerns such as deception, privacy misuse, coercion, manipulation and responsible technology use.
- Provide legal-awareness mapping to relevant Indian legal frameworks without presenting the website as legal advice.
- Create an interactive awareness prototype that helps users recognize red flags and choose safer actions.
- Promote evidence preservation, independent verification and appropriate reporting through official channels.
- Demonstrate responsible design by minimizing personal-data collection and clearly stating the educational purpose of the tool.

Scope

The project is focused on awareness and education. It does not investigate individual suspects, authenticate user identities, process complaints, recover money, or provide professional legal advice. The Scam Detector is an educational risk indicator based on user-selected warning signs. The prototype intentionally avoids storing personal information.

Assignment Alignment

The assignment requires a project report of 10–15 pages, a prototype/creative output, and a 10–12 slide presentation. Its rubric emphasizes problem identification, cyber-threat and legal analysis, innovation, technical quality, documentation, presentation, ethical/social/governance perspective, and authentic references including I4C Cyber Digest material.

2. Methodology

A source-first methodology was followed. The assignment document was used to define the required deliverables and evaluation criteria. Official I4C/National Cyber Crime Reporting Portal advisories were then used to select documented cybercrime patterns. Legal references were taken from India Code and official government sources. Finally, the selected findings were translated into website content and user-facing safety workflows.

Research Workflow

Step	Activity	Purpose
1	Assignment analysis	Identify mandatory sections, deliverables and rubric.
2	Source selection	Prefer official I4C advisories and government legal sources.
3	Case extraction	Record date, modus operandi, red flags and precautions.
4	Ethical analysis	Identify deception, privacy, coercion and responsible-use issues.
5	Legal awareness	Map potentially relevant provisions while preserving factual uncertainty.
6	Prototype design	Convert findings into cards, filters, a detector, checklist and response guide.
7	Validation	Check that the website states its limitations and points victims to official channels.

Design Principles

- No unnecessary personal-data collection.
- Educational explanations instead of automated legal conclusions.
- Clear separation between awareness content and official reporting channels.
- Simple language, progressive disclosure and actionable prevention steps.
- Accessibility-oriented HTML structure, including skip navigation and labelled controls.

3. Selected I4C Case Portfolio

The following five cases were selected from the I4C advisory archive because they represent different social-engineering and financial-fraud patterns and can be translated into practical digital-safety lessons.

Case	Date	Primary Pattern	Key Learning
01	15 Oct 2025	WhatsApp account renting	Never link accounts through unknown QR/pages.
02	31 Oct 2025	Matrimonial investment/crypto	Trust should not replace independent verification.
03	8 Apr 2025	Fake CAPTCHA jobs	Upfront fees + unrealistic earnings are major red flags.
04	6 Mar 2025	Digital arrest	Threats from supposed officials require independent verification.
05	22 Apr 2024	Fake stock investment app/site	Verify investment channels outside social-media groups.

Cross-Case Pattern

Although the scams differ in surface appearance, they repeatedly use a common sequence: establish contact, create credibility or urgency, request a sensitive action, exploit trust or fear, and prevent the victim from independently verifying the claim. CyberSafe therefore teaches a reusable response model: PAUSE → VERIFY → PROTECT → PRESERVE → REPORT.

Source Authenticity

The I4C advisory archive is maintained through the National Cyber Crime Reporting Portal. The selected advisories identify modus operandi and precautions rather than providing invented victim narratives. The project preserves that source-oriented approach.

4. Case Analysis – Cases 01 and 02

Case 1: WhatsApp Web Account Renting Scam

I4C advisory date: 15 October 2025. **Primary classification:** Account misuse / social engineering / secondary cyber fraud.

Attack flow: Social-media earning advertisement → fake page/APK → QR/linking request → WhatsApp account linked → compromised account used for further fraud.

Ethical issues: Deception, misuse of identity/account access, privacy violation, and creation of downstream victims.

Preventive measures: Do not link WhatsApp through unknown QR codes or pages; avoid unknown APKs; verify earning opportunities independently; protect account credentials and recovery settings.

Source: I4C, WhatsApp Web Account Renting Scam – using Facebook & Instagram.

Case 2: Matrimonial Platforms for Investment/Crypto Frauds

I4C advisory date: 31 October 2025. **Primary classification:** Impersonation, relationship-based social engineering, investment/crypto fraud.

Attack flow: Fake profile → rapid trust-building → emotional relationship → investment/crypto proposal → urgency/high-return promise → financial loss.

Ethical issues: Identity deception, exploitation of emotional trust, misuse of personal information, and manipulative financial persuasion.

Preventive measures: Verify identity; use reverse-image checks where appropriate; avoid sharing sensitive information; independently verify investments; reject unrealistic returns.

Source: I4C, Misuse of Matrimonial Platforms for Investment/Crypto Frauds.

4. Case Analysis – Cases 03 and 04

Case 3: Fake CAPTCHA Filling Jobs

I4C advisory date: 8 April 2025. **Primary classification:** Online job fraud / advance-fee fraud / personal-data collection.

Attack flow: High-income job claim → registration/training/software fee → fake work platform → payout blocked → extra taxes/processing fees → scammer disappears.

Ethical issues: False representation, exploitation of job seekers, financial manipulation, and unnecessary collection of personal/financial information.

Preventive measures: Avoid upfront fees for suspicious jobs; research the platform; do not share sensitive information; treat unrealistic earnings as a red flag; report promptly.

Source: I4C, Rise in “Fake CAPTCHA Filling Jobs” Based Cybercrime.

Case 4: Digital Arrest Scam

I4C advisory date: 6 March 2025. **Primary classification:** Impersonation, intimidation, extortion and online economic crime.

Attack flow: Spoofed call → fake law-enforcement identity → accusation → prolonged video-call confinement → demand for money → extortion.

Ethical issues: Abuse of authority imagery, intimidation, coercion, isolation, and exploitation of fear.

Preventive measures: Remember that there is no concept of digital arrest under Indian law; independently verify officials; do not transfer money or disclose sensitive data under pressure; report immediately.

Source: I4C, “Digital Arrest” Scam.

4. Case Analysis – Case 05 and Comparative Findings

Case 5: Fake Stock Market Investment Website/App

I4C advisory date: 22 April 2024. **Primary classification:** Investment fraud / impersonation / social-engineering-enabled financial fraud.

Attack flow: Social-media advertisement → fake expert/group → fake trading app or website → small initial gains/trust → larger deposits → withdrawal problem → loss.

Ethical issues: Misrepresentation, manipulation of trust, deceptive financial claims, and misuse of digital platforms.

Preventive measures: Use verified investment channels; independently validate apps and entities; avoid guaranteed-return claims; do not transfer money based only on social-media groups.

Source: I4C, Beware of Fake Stock Market Investment Website/App.

Comparative Findings

- Trust is manufactured through fake profiles, fake expertise, fake employment offers, fake authority and fake interfaces.
- Urgency reduces the time available for independent verification.
- Financial loss is often preceded by a smaller commitment designed to increase trust.
- Personal information can be both a target and a tool for later manipulation.
- Awareness tools should teach users to recognize patterns rather than depend on one fixed scam script.

5. Ethical and Legal Analysis

Ethical Issues

- Privacy: Scam operations may seek personal, identity, financial or account information beyond what is necessary for legitimate interaction.
- Deception: False identities, fabricated authority, fake employment offers and misleading investment claims undermine informed decision-making.
- Autonomy: Fear, emotional pressure and urgency can interfere with a person's ability to make a voluntary, informed choice.
- Responsible technology use: Digital platforms can be used constructively, but users should verify information, protect credentials and avoid enabling secondary harm.
- Academic and design ethics: CyberSafe avoids collecting personal information and does not claim to replace official authorities.

Legal Awareness Mapping

The Information Technology Act, 2000 contains Section 66C concerning identity theft and Section 66D concerning cheating by personation using a computer resource or communication device. These provisions can be relevant to fact patterns involving misuse of passwords/unique identifiers or online personation; applicability depends on the actual facts.

The Bharatiya Nyaya Sanhita, 2023 contains provisions dealing with cheating and cheating by personation. The project treats these as legal-awareness references rather than automated determinations of guilt.

The Digital Personal Data Protection Act, 2023 establishes a framework for processing digital personal data and recognizes individual interests in protecting personal data. It is relevant to the project's privacy-by-design discussion, while the exact application of the Act depends on the circumstances and applicable commencement/provisions.

Legal Disclaimer

CyberSafe is an academic awareness prototype. It does not provide legal advice, determine whether a criminal provision applies to a particular person, or replace law-enforcement investigation. Users facing an actual cybercrime should use official reporting mechanisms.

6. Proposed Solution – CyberSafe

CyberSafe is designed as an interactive digital-safety portal rather than a passive information page. The prototype converts case research into four user actions: recognize, analyze, protect and report.

Core Modules

Module	Purpose
Case Studies	Explains documented I4C scam patterns and attack flows.
Threat Explorer	Filters threats by financial, identity, social-engineering, shopping and employment risk areas.
Scam Detector	Uses user-selected warning signs to produce an educational risk indicator.
Cyber Law Guide	Introduces selected legal provisions with a fact-dependent disclaimer.
Digital Ethics	Promotes privacy, consent, verification, IP respect, MFA and responsible sharing.
Safety Checklist	Lets users complete a local ten-point cyber-hygiene checklist.
Incident Response	Provides immediate steps: stay calm, contact provider, preserve evidence and report.
Official Help	Directs users to the National Cyber Crime Reporting Portal and 1930.

User Journey

A typical user can begin on the homepage, explore a case study, filter related threats, test a suspicious situation using the Scam Detector, review relevant legal-awareness material, complete the safety checklist, and finally use the official reporting link if a real incident has occurred.

7. Technical Implementation and Expected Outcomes

Technical Implementation

- Frontend: HTML5, CSS3 and Vanilla JavaScript.
- Responsive layout with semantic sections and mobile navigation.
- Client-side rendering of case and threat cards.
- Interactive filters for threat categories.
- Form-based Scam Detector using radio-button inputs and client-side scoring logic.
- Progress-tracked local safety checklist.
- Modal/detail interaction for expanded educational content.
- External official reporting link to cybercrime.gov.in and 1930 telephone link.
- No backend database is required for the awareness prototype; the website states that personal information is not requested, stored or transmitted.

Expected Outcomes

- Improved recognition of common scam patterns.
- Better understanding of how social engineering manipulates trust, fear and urgency.
- Improved awareness of relevant Indian cyber-law concepts.
- Greater awareness of privacy and responsible digital citizenship.
- Clearer knowledge of what evidence to preserve and where to report cybercrime.
- A reusable academic prototype that can later be expanded with verified I4C digest records.

Limitations

- The Scam Detector is an educational indicator, not a professional fraud-detection engine.
- Legal mappings are informational and fact-dependent.
- The prototype does not verify user-submitted claims or investigate suspects.
- The project relies on public advisories and therefore does not expose confidential case information.
- Future versions could connect a curated, verified I4C digest dataset to the interface.

8. Conclusion

CyberSafe demonstrates how ethical and legal education can be embedded into a practical digital product. Instead of treating cybercrime as a purely technical problem, the project connects attack mechanisms with human behaviour, privacy, ethics, legal awareness and responsible reporting.

The five I4C-based case studies show that many cyber scams depend on recurring psychological and digital patterns: manufactured trust, impersonation, urgency, unrealistic promises, fake interfaces and pressure to act before verification. The prototype translates these patterns into interactive learning modules so that users can practice safer decision-making.

The project also follows responsible-technology principles by minimizing personal-data collection, clearly identifying itself as an academic project, avoiding claims of official authority, and directing victims to official reporting channels. The result is a feasible educational prototype that satisfies the assignment's focus on innovation, technical implementation, cyber-threat analysis, ethics, law, documentation and presentation.

Future Enhancements

- Curated I4C Daily Digest database with source/date metadata and search.
- Case comparison dashboard and trend visualization.
- Multilingual awareness content for wider accessibility.
- Scenario-based quizzes and score history stored only with explicit consent.
- Accessibility improvements including keyboard-first interaction and additional screen-reader testing.
- Teacher/admin mode for classroom demonstrations without collecting student-sensitive information.

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